

TrueGrain Deck™ Laminate Construction

Five-Layer PMMA Film, Engineered by RENOLIT

The TrueGrain decorative surface is a precision-laminated, five-layer PMMA film stack engineered by RENOLIT and bonded to a domestically extruded cellular PVC substrate at Patwin Plastics in Linden, New Jersey. Each layer performs a distinct functional role, from abrasion resistance on the wear face to chemical adhesion at the substrate interface. The diagram and descriptions below are reproduced without alteration and are intended for use by architects, specifiers, dealers, and distributor partners.

● ENGINEERED BY RENOLIT

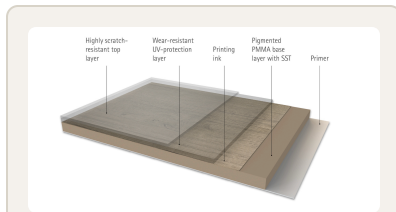


FIGURE 1.
TRUEGRAIN LAMINATE
CROSS-SECTION. LAYERS
SHOWN IN TRUE STACKING
ORDER, TOP TO BOTTOM.

PRODUCT	TrueGrain Deck™ capped cellular PVC
SURFACE	RENOLIT exterior PMMA laminate
FILM CHEMISTRY	PMMA (acrylic) with SST
SUBSTRATE	Cellular PVC with ASA capstock
BONDING	Primer-assisted permanent lamination
ORIGIN	Laminated and extruded in Linden, NJ, USA
WARRANTY	25-year limited residential

01 Highly scratch-resistant top layer SURFACE DURABILITY

Clear, high-hardness exterior surface film engineered to resist abrasion from foot traffic, patio furniture, pet claws, sand, and grit. Defines the wear face of the board and protects the printed decor against everyday mechanical contact in residential service.

02 Wear-resistant UV-protection layer WEATHERING & COLOR HOLD

Dedicated UV stabilizer film formulated to absorb solar radiation before it can reach the print or the polymer substrate. Preserves color fidelity on the decorative wood-grain and shields the underlying chemistry from photo-oxidative degradation.

03 Printing ink PHOTOREALISTIC DECOR

High-resolution rotogravure decor print laid down between the protective films above and the pigmented PMMA base below. Reproduces variegated grain, pore structure, and tonal depth of premium hardwoods with edge-to-edge registration on every board.

04 Pigmented PMMA base layer with SST HEAT REFLECTION & FILM BACKBONE

Acrylic (PMMA) base layer carrying RENOLIT's Solar Shield Technology (SST). SST reflects the near-infrared portion of the solar spectrum so the board runs measurably cooler under direct sun, while the PMMA chemistry provides the dimensional stability and weatherability specified for exterior cladding.

05 Primer SUBSTRATE ADHESION

Engineered tie-coat applied to the underside of the film stack. Forms the chemical bond that locks the laminate to the capped cellular PVC substrate, producing a permanent lamination rather than a surface-applied coating.

RENOLIT is a leading global manufacturer of high-performance polymer films. Their exterior PMMA foils, including the Solar Shield Technology shown above, are specified in window, door, and architectural cladding products worldwide. American Pro Building Products laminates the RENOLIT TrueGrain film to a domestically extruded cellular PVC substrate, marrying European film science to an American-made deck board.

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